BOM Creator Agent v9.2.0 - Production Context Report

Date: 2026-02-18
Project: amb_w_tds v9.2.0
Branch: feature/v9.2.0-development
Author: Matrix Agent

Executive Summary

This report documents the real-world production data structure for AMB-Wellness Aloe Vera products. The BOM Creator Agent must be adapted to handle this multi-level, fractional-quantity production environment.

---

1. Product Hierarchy

```
┌──────────────────────────────────────────────────────────────────────┐
│                    AMB-Wellness Production Chain                     │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│   M033 (Raw Material)                                                │
│   └── Aloe Vera Gel (Fresh Leaves)                                   │
│        │                                                             │
│        ▼                                                             │
│   0227 (Sub-Assembly - Liquid)                                       │
│   └── INNOVALOE ALOE VERA GEL CONCENTRATE 30:1                       │
│   └── Concentration: 30X | Brix: 10.0 | UOM: Kg                      │
│   └── Has Variants: YES (template item)                              │
│        │                                                             │
│        ▼                                                             │
│   0307 (Finished Product - Powder)                                   │
│   └── INNOVALOE ALOE VERA GEL SPRAY DRIED POWDER 200:1               │
│   └── Concentration: 200X | Brix: 95.0 | UOM: Kg                     │
│   └── Has Variants: NO                                               │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

---

2. Key Items Reference

2.1 Item 0307 - Finished Product (Powder)

Field	Value
Item Code	0307
Item Name	INNOVALOE ALOE VERA GEL SPRAY DRIED POWDER 200:1
Item Group	Products Powder
Stock UOM	Kg
Concentration	200X
Brix Level	95.0
Has Variants	No
Default BOM	BOM-0307-004
Shelf Life	730 days
Quality Template	STD Inspection Template Powder 200X

Item Attributes:
- Aloin Content
- Arsenic
- Polysaccharide
- Molecular Weight
- Special Properties 1
- Packing Type Powder 1/2/3
- Microbiological

2.2 Item 0227 - Sub-Assembly (Concentrated Juice)

Field	Value
Item Code	0227
Item Name	INNOVALOE ALOE VERA GEL CONCENTRATE 30:1
Item Group	Products Liquid
Stock UOM	Kg
Concentration	30X

Field	Value
Brix Level	10.0
Has Variants	Yes (Template Item)
Shelf Life	365 days
Quality Template	STD Inspection Template Liquid Concentrated

Item Attributes:
- Liquid Concentration
- Special Properties Certificate 1
- Juice Type
- Preservatives
- Processing Method
- Special Properties 1
- Packing Type
- UF (Ultra-Filtered)
- Hidrolyzed
- Raspaduras

---

3. BOM Structure Analysis (BOM-0307-004)

3.1 Overview

- **BOM Name:** BOM-0307-004
- **Output Item:** 0307 (Powder 200:1)
- **Quantity:** 1 Kg
- **UOM:** Kg
- **Total Cost:** 517,924.06 MXN
- **Status:** Submitted (docstatus=1), Active, Default

3.2 BOM Components Breakdown

Primary Components

Item Code	Item Name	Qty	UOM	Rate (MXN)	Amount
0227	Concentrate 30:1	1.0	Kg	150.00	150.00
M033	Aloe Vera Gel	0.083607	Kg	1.40	0.12

Container (CRITICAL - Fractional Piece)

Item Code	Item Name	Qty	UOM	Rate (MXN)	Amount
E001	**220L Barrel Blue**	**0.004351**	**Piece**	787.00	3.42

Filter Aids & Process Materials

Item Code	Item Name	Qty	UOM	Rate (MXN)
M004	Diatomaceous Earth Celite 512	0.062551	Kg	14.88
M005	Diatomaceous Earth Celite 501	0.021398	Kg	14.88
M006	Diatomaceous Earth Creaclar SC150	0.009789	Kg	72.80
M042	Activated Carbon Ultra	0.198684	Kg	61.00
M045	ENZIMA ROHAPECT	0.00435	Kg	592.26

Utilities

Item Code	Item Name	Qty	UOM	Rate (MXN)
WATER	Process Water	500	Litre	547.69
ELECTRIC	Electricity	150	kWh	211.48
GAS	Natural Gas	50	Gigajoule	262.22
LABOR	Direct Labor	8	Hour	24,868.10

Services & Overhead

Item Code	Item Name	Qty	UOM
0227-Utility	Utility Services	1.0	Service
0227-Supplies-Material	Supplies	1.0	Kg
0227-Packing-Material	Packing	1.0	Piece
TRANSP_LEAF	Leaf Transport	0.01076	Service
AN_LABORA	Lab Analysis	0.00099	Unit
MANT_PLAN	Plant Maintenance	0.01076	Unit

4. Critical Business Rules

4.1 Container Handling (IMPORTANT)

```
┌──────────────────────────────────────────────────────────────────┐
│  CONTAINERS ARE ITEMS, NOT UOMs                                  │
├──────────────────────────────────────────────────────────────────┤
│  ❌ WRONG: UOM = "IBC" or UOM = "Drum"                           │
│  ✅ RIGHT: Item = "E001" (220L Barrel) with UOM = "Piece"        │
│  ✅ RIGHT: Item = "E011" (IBC Container) with UOM = "Piece"      │
└──────────────────────────────────────────────────────────────────┘
```

Container Items:

Item Code	Description	Capacity
E001	220L Barrel Blue	220L (~238 Kg)
E011	IBC Container	1000L (~1080 Kg)
E012	Reused IBC Container	1000L (~1080 Kg)

4.2 Fractional Quantities

The BOM uses fractional container quantities to represent usage per Kg of output:

```
Container Qty = 1 / Container Capacity (Kg)

Example (E001 - 220L Barrel):
- Capacity: ~230 Kg
- Qty per 1 Kg output: 0.004351 Piece
- For 2000 Kg production: 2000 × 0.004351 = 8.7 Barrels ≈ 9 Barrels
```

4.3 Equivalent Powder Measurement

Liquid products (0227) are measured by their **equivalent powder weight**, not actual liquid weight:

```
Example:
- 2000 Kg equivalent powder (0303)
- = 18,000 Kg concentrated juice (0227)
- = 36,000 Kg fresh aloe leaf (M033)
```

4.4 UOM Standards

Product Type	UOM
Powder Products	Kg
Liquid Concentrates	Kg
Raw Materials	Kg
Containers	Piece
Utilities (Water)	Litre / Cubic Meter
Utilities (Electric)	kWh
Utilities (Gas)	Gigajoule
Labor	Hour
Services	Service / Unit

---

5. TDS (Technical Data Sheet) Parameters

5.1 Quality Parameters for Powder (0307)

- pH
- Ash Content
- Polysaccharides
- Aloin Content
- Arsenic Levels
- Molecular Weight
- Microbiological Tests

5.2 Quality Parameters for Concentrate (0227)

- Brix Level
- pH
- Polysaccharides
- Color
- Turbidity

---

6. API Endpoints

6.1 Sandbox Environment

```
Base URL: https://sysmayal.ngrok.io
Auth: token 3eb0e3d4376d8b0:0c832e78bc5e537
```

6.2 Key Endpoints

```
# Get Item
GET /api/resource/Item/{item_code}

# Get BOM
GET /api/resource/BOM/{bom_name}

# Create BOM (via Agent)
POST /api/method/
amb_w_tds.ai_bom_agent.api.create_multi_level_bom_from_spec
Body: {"request_text": "...", "dry_run": false}
```

---

7. Agent Implementation Status

7.1 Current Fixes Applied

Commit	Fix Description
923fa36	Convert BOMItem dataclass to dict before create_bom call
ed66d07	Fix IBC UOM issue - IBC is an Item not a UOM
0a55b32	Update Item Group to SFG Semi Finished Goods
ea588eb	Add mandatory product_key and mx_product_service_key defaults
f427ebb	Python boolean False fix

7.2 Parser Configuration (parser.py)

```
PACKAGING_FORMATS = {
    "1000L-IBC": {
        "uom": "Kg",
        "volume_liters": 1000,
        "capacity_kg": 1080,
        "container_item": "E011"
    },
    "200L-DRUM": {
        "uom": "Kg",
        "volume_liters": 200,
        "capacity_kg": 216,
        "container_item": "E001"
    },
    "20L-PAIL": {
        "uom": "Kg",
        "volume_liters": 20,
        "capacity_kg": 21.6,
        "container_item": "E001"  # Needs real code
    }
}
```

---

8. Next Steps

8.1 Immediate (Phase 3 Completion)

1. ✅ Fix BOMItem to dict conversion (923fa36)
2. ⏳ Re-run Phase 3 tests on sandbox
3. ⏳ Verify all T3.1-T3.8 tests pass

8.2 Phase 4 - Hardening

1. Feature flags for production rollout
2. Comprehensive logging
3. Rollback capabilities
4. Documentation updates

8.3 Future Enhancements

1. TDS parameter validation integration
2. Multi-level BOM explosion for 0227 sub-assembly
3. Variant handling for template items
4. Cost calculation integration

---

9. Appendix: Test Commands

Pull Latest & Test

```
cd ~/frappe-bench/apps/amb_w_tds
git pull origin feature/v9.2.0-development
bench --site v2.sysmayal.cloud clear-cache

# Run BOM Creation Test
bench --site v2.sysmayal.cloud execute \
  amb_w_tds.ai_bom_agent.api.create_multi_level_bom_from_spec \
  --kwargs '{"request_text": "0227-ORGANIC-VANILLA-1000L-IBC",
"dry_run": false}'
```

Full Phase 3 Test Suite

```
bash scripts/test_v9.2.0_phase3_mrp.sh v2.sysmayal.cloud
```

Report generated for AMB-Wellness Orchestrator Team